InsightPilot: Evaluation and CI Pipeline

Test levels on every pull request · evaluation harness · release · periodic human evaluation

EVERY PULL REQUEST: GITHUB ACTIONS (ALL JOBS MUST PASS TO MERGE)

Static analysis
Lint, format, strict types
Python and TypeScript
ruff · mypy · tsc · eslint

→

Unit and contract
Plan schema and validation
StepResult, connector contract
pytest

→

Integration
Real PostgreSQL 16 and DuckDB
Read-only enforced; audit written
pytest · Docker Compose

→

Agent end-to-end
Plan → agents → verify → hop
Recorded model responses
pytest · model fixtures

→

UI end-to-end
Ask, approve, inspect, sign off
Accessibility checks
Playwright · axe-core

→

Security
Secrets, dependencies, images
Every write attempt must fail
gitleaks · scanners

harness job runs on the same pull request ↓

↑ failing test or regression blocks the merge

EVALUATION HARNESS: ON EVERY PULL REQUEST AND NIGHTLY

Question and defect sets
30–50 questions, ground truth
8 seeded-defect classes (D1–D8)
YAML + Parquet fixtures

→

Harness runner
Replay on PRs (recorded responses)
Live nightly: 5 trials per question
pytest · Docker Compose

→

System under test
Full pipeline, frontier model
Verifier on and off (B1)
InsightPilot

→

Scoring
Accuracy · silent errors · coverage
False alarms · pass^k (k = 1–5) · cost
Python · bootstrap CIs

→

Gate and report
Fails if accuracy, coverage
or false-alarm rate regress
GitHub Actions

all jobs green: merge and release ↓

↑ release candidate under test

RELEASE AND DEPLOYMENT: SINGLE-TENANT, SELF-HOSTED

Merge to main
Reviewed pull request
All jobs and harness green
GitHub

→

Build and tag images
Versioned, pinned images
Backup before migrations
Docker

→

One-command quickstart
docker compose up + demo data
Clone to first answer < 30 min
Docker Compose

→

Expo demo deployment
One organisation per stack
Seeded demo schema
Docker Compose

release used for studies (Dec 7–11) ↓

↑ findings become backlog items

PERIODIC HUMAN EVALUATION: NOT IN CI

Verification-cost study
Five analysts, timed sessions
Check vs derive (NFR-10)
moderated sessions

→

Seeded-defect review
Reject planted defects
With and without trace (NFR-09)
moderated sessions

→

Usability walkthrough
Ask, approve, inspect, sign off
Findings fed into the backlog
think-aloud sessions

KEY: Solid arrow = the run proceeds; dashed arrow = feedback. Cards run left to right within a lane. B1 = verifier-off baseline (Table 12).

Requirements: FR-01, FR-02 and NFR-14 (harness in CI on every pull request); NFR-13 and FR-27 (quickstart); NFR-09 and NFR-10 (human studies).